

IT NETWORK DOCUMENTATION MANUAL

Phase 08: Documentation

Network Reference, Operations & Change Record

Document ID: P07-DOC-NET-001 | **Version:** 1.0 | **Owner:** IT Support / IT Infrastructure | **Classification:** Internal Use

1. Purpose and Scope

This manual establishes the standard format for documenting the company's logical and physical network environment. It is intended for IT Support, Infrastructure, Network Administration, and authorized technical personnel. Production values must be maintained in an approved controlled repository and must never include passwords, private keys, shared secrets, or other authentication material.

2. Documentation Principles

- Keep network information accurate, current, and attributable to an approved change or source.
- Document both logical relationships and relevant physical infrastructure.
- Use consistent naming for sites, devices, interfaces, VLANs, subnets, and services.
- Record changes with date, owner, approval, and validation information.
- Separate general network documentation from confidential credentials and secrets.
- Sanitize network diagrams and screenshots before using them in training or public-facing materials.

3. Network Documentation Scope

Area	Documentation Required
Sites	Site name, location, business function, and connectivity type
WAN/Internet	Provider, circuit reference, edge device, and approved service details
Firewall/Router	Hostname, role, interfaces, zones, and approved configuration reference
Switching	Device ID, role, location, uplinks, VLANs, and management reference
Wireless	AP identifiers, locations, SSIDs, controller/service reference, and coverage notes
IP Addressing	Subnet, gateway, DHCP range, DNS, VLAN, purpose, and ownership
Servers/Services	Service name, network segment, dependencies, and owner

Monitoring	Approved monitoring and alerting references
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4. High-Level Logical Architecture

Standard reference flow:

Internet/WAN → Edge Firewall/Router → Core/Distribution Layer → Access Switching → Endpoints / Servers / Wireless Access Points

Actual production topology must be maintained in the organization's approved network diagram. This document intentionally uses generic architecture and does not contain real infrastructure addresses or credentials.

5. Network Inventory Template

Device ID	Hostname	Type	Role	Location	Management IP	Status	Owner
NET-001	[hostname]	Firewall	Edge	[site]	[approved IP]	Active	Network Team
NET-002	[hostname]	Switch	Core/Distribution	[site]	[approved IP]	Active	Network Team
NET-003	[hostname]	Switch	Access	[site]	[approved IP]	Active	Network Team
NET-004	[hostname]	Wireless AP	Access	[site]	[approved IP]	Active	Network Team

6. IP Addressing Documentation

Network/VLAN	Subnet	Gateway	DHCP Range	DNS	Purpose	Owner
VLAN-[ID]	[CIDR]	[Gateway]	[Range]	[DNS]	Corporate users	Network Team
VLAN-[ID]	[CIDR]	[Gateway]	[Range]	[DNS]	Servers	Infrastructure
VLAN-[ID]	[CIDR]	[Gateway]	[Range]	[DNS]	Guest/isolated	Network Team

Replace placeholders only with approved production values. Do not place credentials, SNMP community strings, API keys, VPN secrets, or private keys in this table.

7. VLAN Documentation

VLAN ID	Name	Purpose	Subnet	Gateway	DHCP	Security Zone	Owner
[ID]	[Name]	Corporate users	[CIDR]	[Gateway]	Yes/No	[Zone]	[Owner]
[ID]	[Name]	Servers	[CIDR]	[Gateway]	Yes/No	[Zone]	[Owner]
[ID]	[Name]	Guest	[CIDR]	[Gateway]	Yes/No	[Zone]	[Owner]

8. Wireless Documentation

AP ID	Location	SSID	Network/VLAN	Status	Controller/Service
AP-[ID]	[Location]	[Approved SSID]	[VLAN]	Active	[Controller]
AP-[ID]	[Location]	[Approved SSID]	[VLAN]	Active	[Controller]

- Use approved encryption and authentication.
- Do not document wireless passwords in general network documentation.
- Record coverage issues, AP replacements, and configuration changes through the approved change process.

9. Firewall and Edge Documentation

Component	Purpose	Interface/Zone	Upstream/Downstream	Owner	Change Reference
[Firewall]	Internet edge	WAN	ISP	Network Team	[Change ID]
[Firewall]	Internal segmentation	LAN/DMZ	Core	Network Team	[Change ID]

Firewall Documentation Rules

- Document rule purpose and business owner at an appropriate level.
- Use approved change references for rule additions or modifications.
- Do not store administrator passwords or secrets in this manual.
- Review unused or obsolete rules according to the organization's firewall-review process.

10. Physical Network Documentation

Maintain an approved physical diagram showing, where applicable, racks, patch panels, switches, uplinks, firewall/router equipment, wireless infrastructure, and key server/network locations.

Location	Rack	Device	Uplink/Port	Power Source	Notes
[Site]	[Rack]	[Device]	[Port]	[PDU/UPS]	[Notes]
[Site]	[Rack]	[Device]	[Port]	[PDU/UPS]	[Notes]

11. Network Services

Service	Purpose	Primary System	Dependencies	Owner	Monitoring
DHCP	Endpoint addressing	[System]	Network/DNS	Infrastructure	[Monitor]
DNS	Name resolution	[System]	Network/Directory	Infrastructure	[Monitor]
NTP	Time synchronization	[System]	Network/WAN	Infrastructure	[Monitor]
Directory/Identity	Authentication	[System]	Network/DNS	IT Administration	[Monitor]

12. Standard Network Troubleshooting Workflow

1. Define the scope: one endpoint, one segment, one site, or widespread.
2. Confirm the affected device and user.
3. Check physical or wireless connectivity.
4. Review endpoint IP configuration.
5. Test local gateway connectivity where authorized.
6. Test an approved internal destination.
7. Test DNS resolution when relevant.
8. Check whether a network service or infrastructure component is unavailable.
9. Document the results and escalate infrastructure issues with evidence.

13. Example Endpoint Checks

Purpose	Example Command/Check
Linux IP configuration	ip addr
Linux routing	ip route
Linux DNS configuration	resolvectl status (where supported)
Connectivity test	ping <approved-destination>
Path test	tracert/route <approved-destination> or approved equivalent
DNS test	nslookup <approved-name> or approved equivalent

Commands should be run only on authorized systems and against approved destinations. Replace placeholders with actual authorized values.

14. Network Incident Documentation

Field	Required Information
Incident ID	Support/ticket reference
Start Time	When the issue began or was reported
Scope	Users/devices/sites affected
Symptoms	Exact observed behavior
Tests	Connectivity/configuration checks performed
Findings	Evidence-based observations
Action	Corrective action or escalation
Verification	How service restoration was confirmed
Owner	Responsible support team

15. Network Change Documentation

Field	Description
Change ID	Approved change/reference number
Requester	Person/team requesting the change
Business Reason	Why the change is required
Affected Devices	Systems/segments/interfaces affected
Planned Date/Time	Approved implementation window

Implementation Plan	Step-by-step approved action
Validation Plan	How success will be confirmed
Rollback Plan	How the change will be reversed if required
Result	Successful / Failed / Rolled Back
Approver	Authorized approver

16. Network Change Procedure

1. Create the change request with business justification.
2. Identify affected services, devices, and users.
3. Assess risk and expected impact.
4. Document the implementation and rollback plan.
5. Obtain required approval.
6. Perform the change during the approved window.
7. Validate connectivity and affected services.
8. Record the final configuration state and outcome.
9. Close the change with evidence and follow-up actions.

17. Monitoring and Health Checks

- Monitor availability of critical network devices and services.
- Review alerts for repeated interface, connectivity, or service failures.
- Track capacity and utilization where monitoring is available.
- Investigate recurring alerts rather than repeatedly dismissing them.
- Document planned maintenance windows and known outages.

18. Network Security Documentation Rules

- Never store passwords, private keys, API tokens, VPN secrets, or similar credentials in this document.
- Do not publish internal network diagrams or addressing information externally without authorization.
- Restrict detailed topology and management information to authorized personnel.
- Sanitize screenshots before using them in training or portfolio documentation.
- Record security-sensitive configuration changes through approved change management.

19. Documentation Review Checklist

Review Item	Complete
Network inventory current	☐
IP/VLAN documentation current	☐
Logical topology current	☐
Physical topology current	☐
Wireless inventory current	☐
Critical network services documented	☐

Change references recorded	☐
Obsolete information removed	☐
No credentials/secrets included	☐
Document owner/review date current	☐

20. Document Maintenance

The IT Infrastructure/Network Team should review network documentation at least quarterly for accuracy and after every significant topology, addressing, device, service, or connectivity change. Production diagrams and addressing records should be stored in an approved controlled repository.

21. Revision History

Version	Date	Description	Owner
1.0	2026-08-10	Initial professional network documentation manual for Phase 07.	IT Support / IT Infrastructure